

L8: Within-module Degree

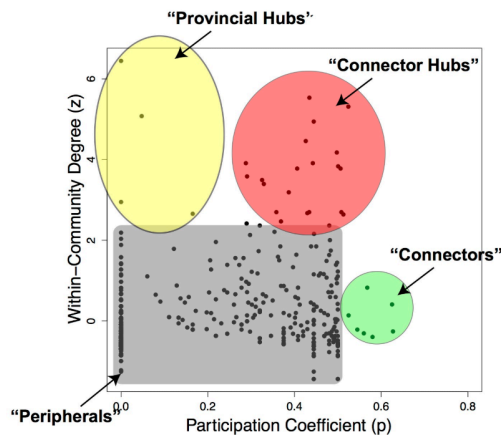


Figure from [Worked Example: Centrality and Community Structure of US Air Transportation Network](https://dshizuka.github.io/networkanalysis/example_usairports.html)

(https://dshizuka.github.io/networkanalysis/example_usairports.html) by Dai Shizuka

Another important question is whether a node is a hub or not within its own community.

A metric to answer this question is the normalized “*within-module degree*”:

$$z_i = \frac{\kappa_i - \bar{\kappa}_i}{\sigma_{\kappa_i}}$$

where κ_i is the “internal” degree of node i , considering only connections between that node and other nodes in the same community, $\bar{\kappa}_i$ is the average internal degree of all nodes that belong in the community of node i , and σ_{κ_i} is the standard deviation of the internal degree of those nodes.

So, if a node i has a large internal degree, relative to the degree of other nodes in its own community, we consider it a *hub within its module*.

If we combine the information provided by both the participation coefficient and the within-community degree, as shown in the visualization, we can identify four types of nodes, based on the nodes that have unusually high values of P_i and z_i :

1. **Connector Hubs:** nodes with high values of both P_i and z_i . These are hubs within their own community that are also highly connected to other communities, forming bridges to those communities.
2. **Provincial Hubs:** nodes with high value of z_i but low value of P_i . These are hubs within their own community but not well connected to nodes of other communities.
3. **Connectors:** nodes with high value of P_i but low value of z_i . These are not hubs within their module – but they form bridges to other communities.
4. **Peripheral nodes:** nodes with low values of both P_i and z_i . Everything else -- this is how the bulk of the nodes will be classified as.

Clearly, these four types are heuristically defined and there are always nodes that would fall somewhere between these four roles.